

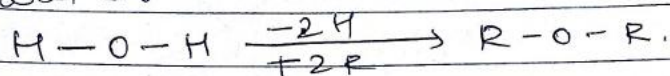
Chapter - 5

Ether

Aliphatic Ether

→ Ethers are organic compound with divalent oxygen atom bonded to two alkyl groups.

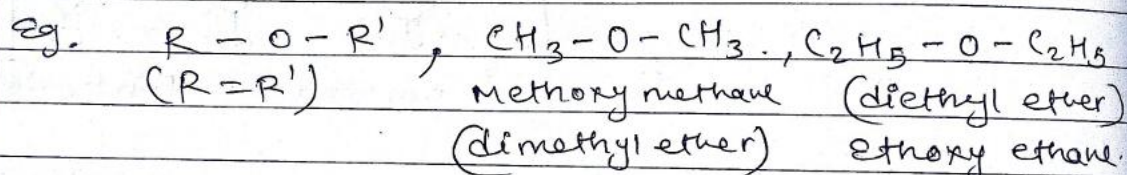
→ They are considered as dialkyl derivative of water.



Classification of ether, + Nomenclature

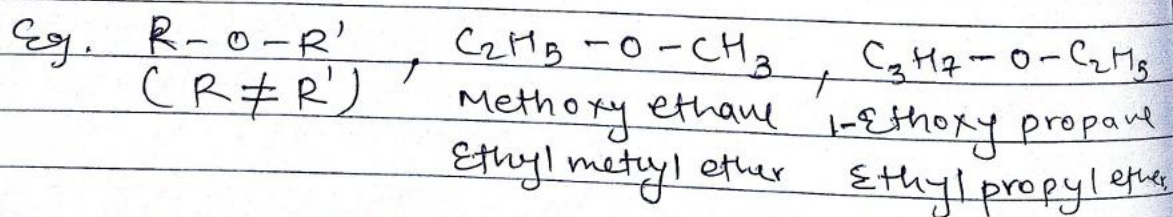
i) Simple / Symmetrical ether

→ Two alkyl groups joined with $(\text{O}-\text{O}-)$ are same.



ii) Unsymmetrical / mixed ether.

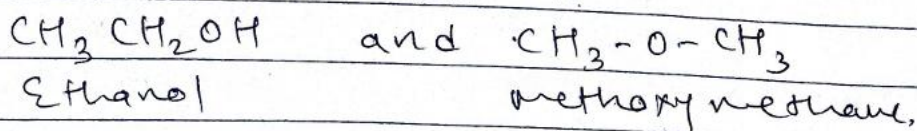
→ Two alkyl group bonded with $(-\text{O}-)$ are different.



Isomerism in Ether

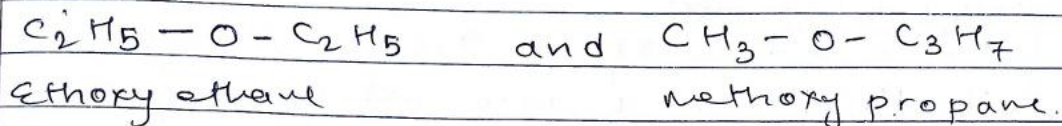
→ They show functional isomerism and metamerism.

① Functional isomerism.



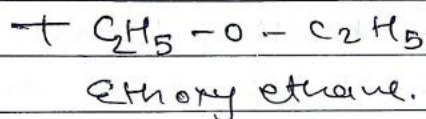
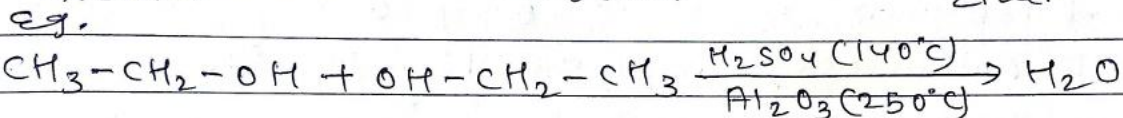
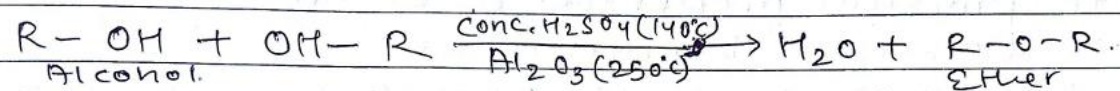
(i) Metamers.

→ Same molecular formula but unequal distribution of alkyl group attached to (-O-).



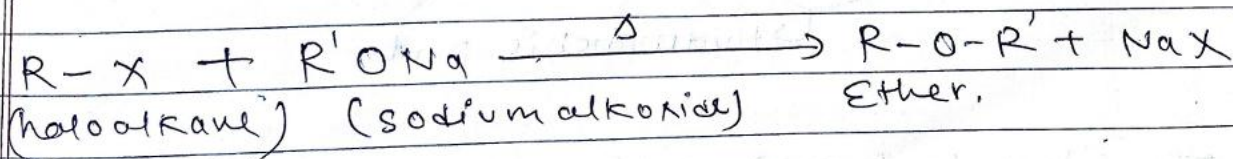
Method of preparation of Ether

(1) From alcohol.

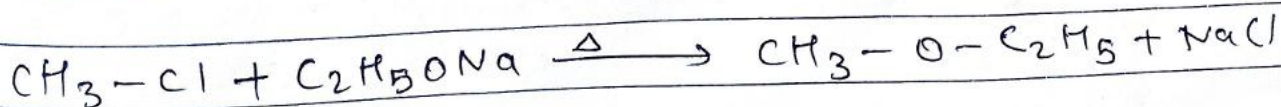
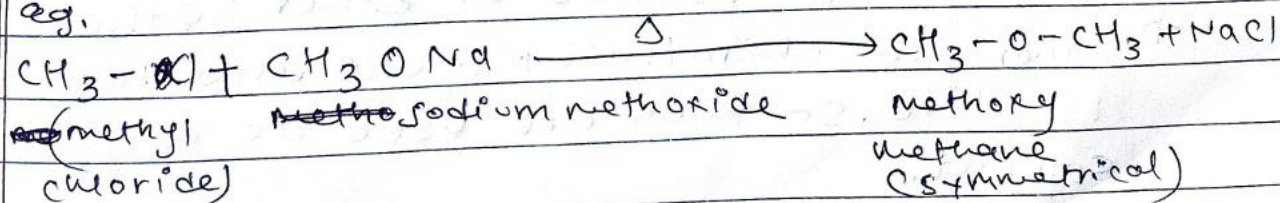


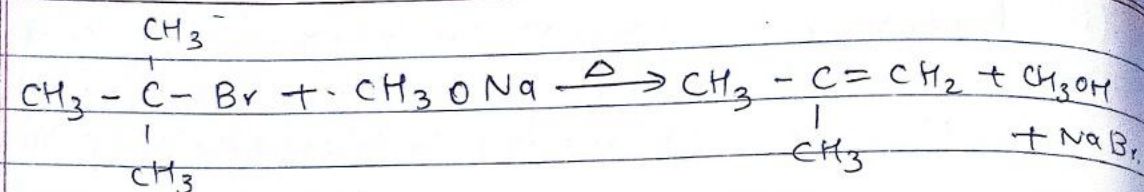
→ Only symmetrical ether can be produced by this method.

(2) From williamson's synthesis

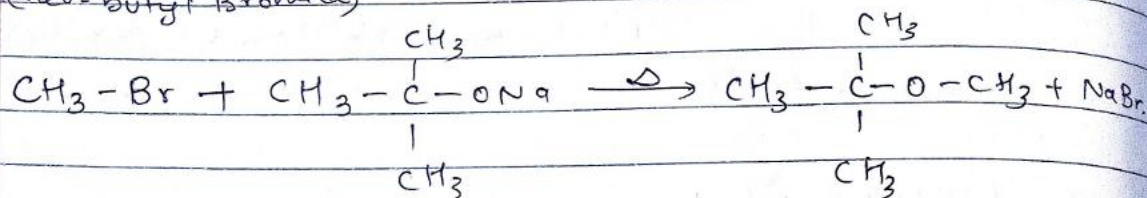


eg.





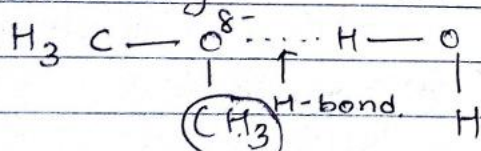
(neo-butyl Bromide)



(Sodium tert. butoxide)

Physical properties of ether.

- Lower members are gases
- Higher " " " liquid.
- Colourless, sweet smelling liquid.
- Highly Flammable.
- Low boiling point as compared to isomeric alcohol.
- Lower members are soluble in water because they can form intermolecular hydrogen bonding with water.



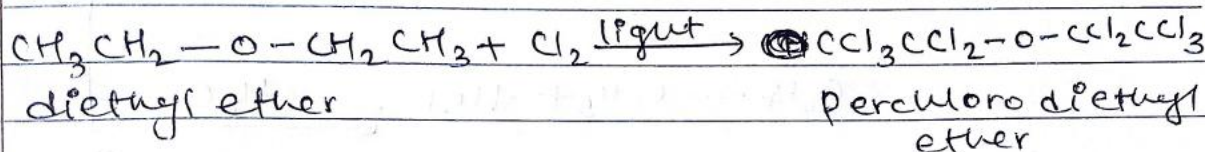
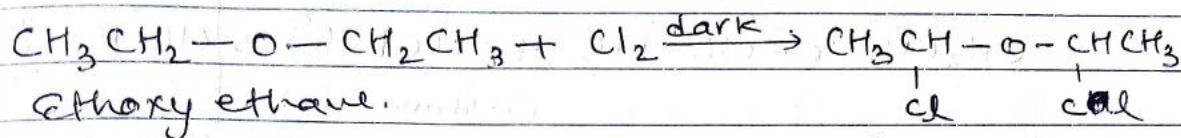
↳ Hydrophobic part.

Chemical properties of Ether [Ethoxyethane].

- Ether are chemically inert and less reactive, so they are used as organic solvent.

(1) Reactions involving alkyl radical's substitution reaction

(*) Halogenation (Chlorination).



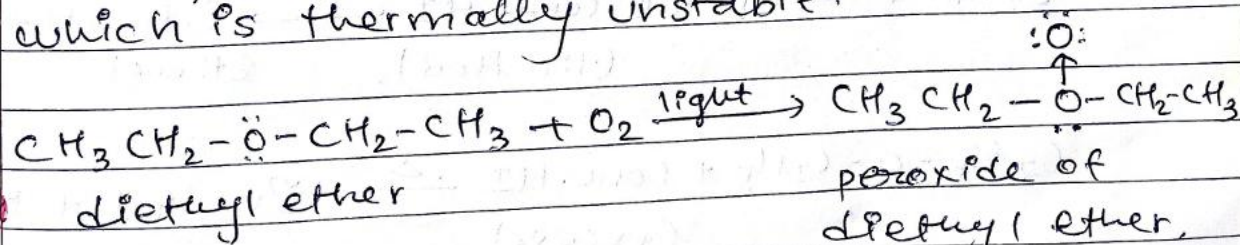
→ When halogenation occurs in dark, only one hydrogen bonded with carbon near oxygen is replaced by halogen.

→ When halogenation occurs in light, all hydrogen bonded with all carbon are replaced by halogen and rxn may be explosive.

2) Rxns involving ethereal oxygen giving as Addition Reaction.

(*) Rxn with air (O₂)

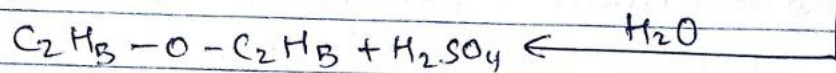
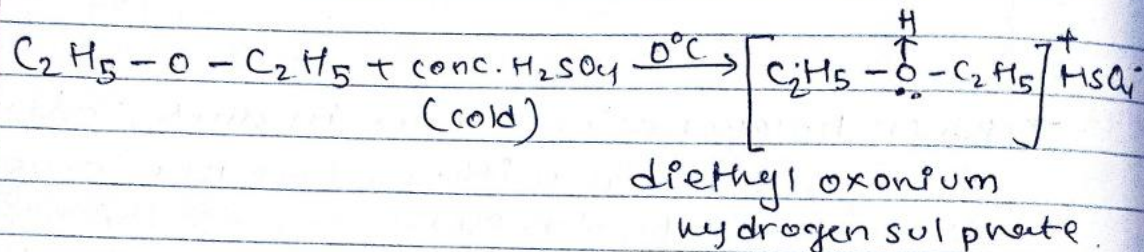
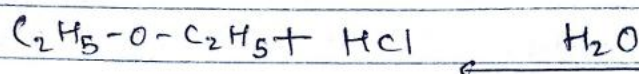
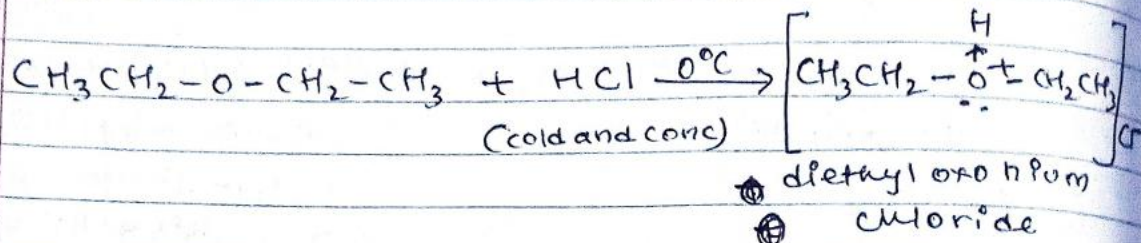
→ When ethoxy ethane (ether) reacts with atmospheric air, ether peroxide is formed which is thermally unstable.



→ To prevent formation of peroxide, ether are stored in dark brown bottle along with iron powder / ethanol and filled upto brim.

ii) Rxn with cold H_2SO_4 and HCl .

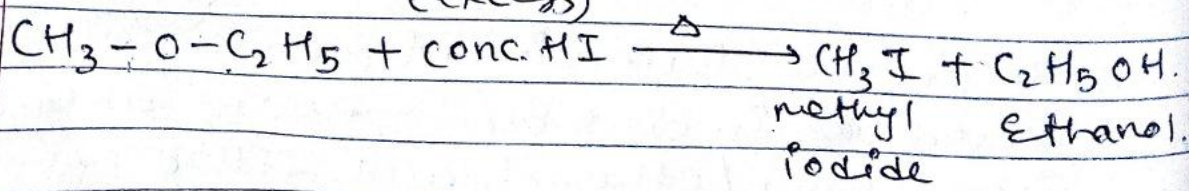
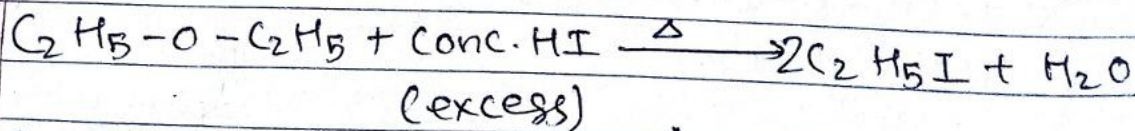
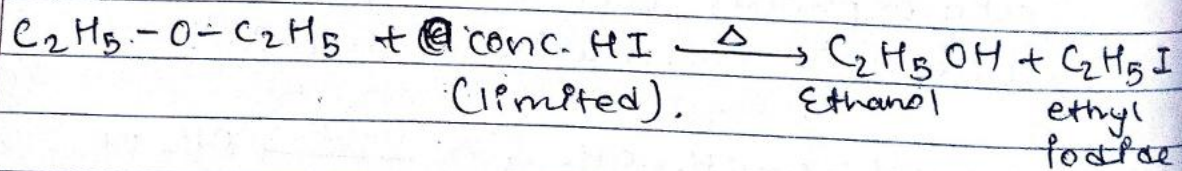
→ Ethoxy ethane (ether) reacts with cold H_2SO_4 and HCl to form oxonium ion.



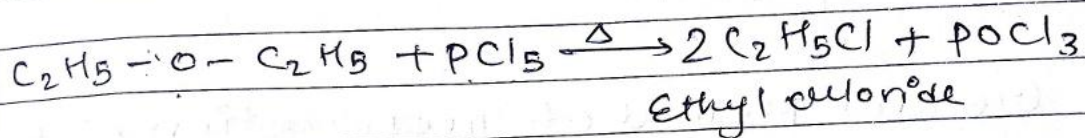
3) Rxn involving breaking of C-O bond.

i) Rxn with HI .

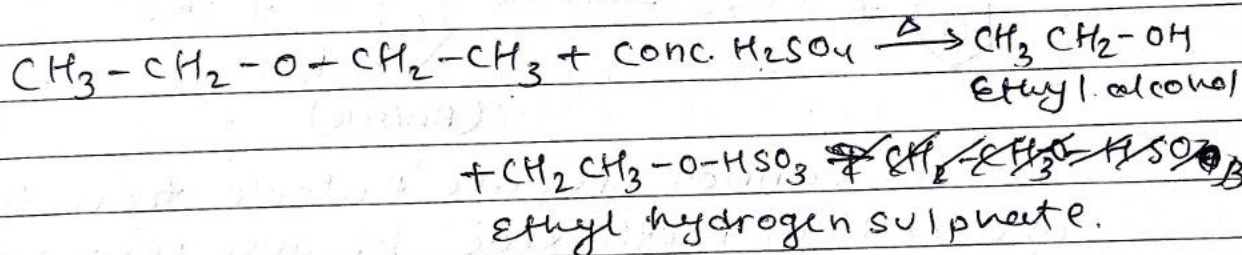
→ Ether reacts with hydroiodic acid to form alcohol and alkyl halide.



- ii) Rxn with PCl_5 .
 $\rightarrow$ When ether is heated with PCl_5 , alkyl chloride is formed.



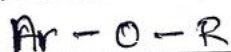
- iii) Rxn with hot and conc. ~~acid~~ H_2SO_4 .
 $\rightarrow$ When H_2SO_4 is heated with ether, ~~alkyl alcohol~~ alkyl alcohol and alkyl hydrogen sulphate is formed.



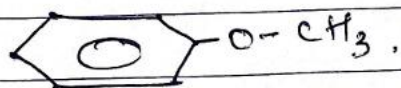
Uses of ether:-

- i) Used as anesthetic in surgery.
- ii) Used as refrigerant
- iii) Used as organic solvent.
- iv) Used in perfumes.

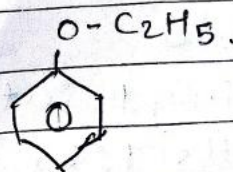
Aromatic Ether (Anisole).



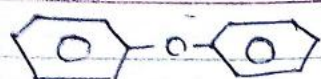
where Ar = Arene  (phenyl)



Methoxy benzene.
(Anisole)



Ethoxy benzene.
(phenetole)

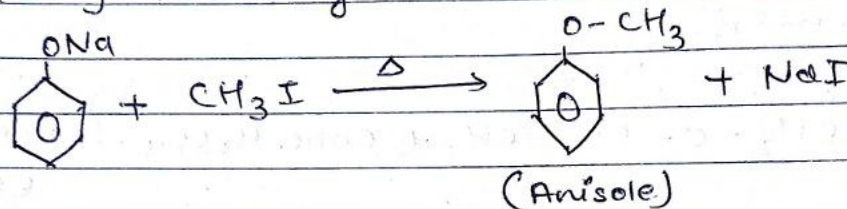


phenoxy benzene
(diphenyl ether)

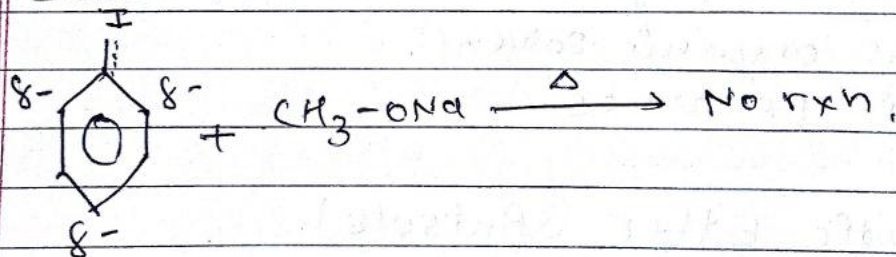
General Method of preparation of Anisole.

i) Williamson's synthesis.

→ Sodium phenoxide when heated with methyl iodide gives Anisole.



but we cannot prepare Anisole from iodobenzene and Sodium Methoxide because resonating structure of Iodobenzene shows partial double bonding between phenyl group and iodine. So, it is hard to replace CH_3 with Iodine.



physical properties of Anisole.

→ colourless

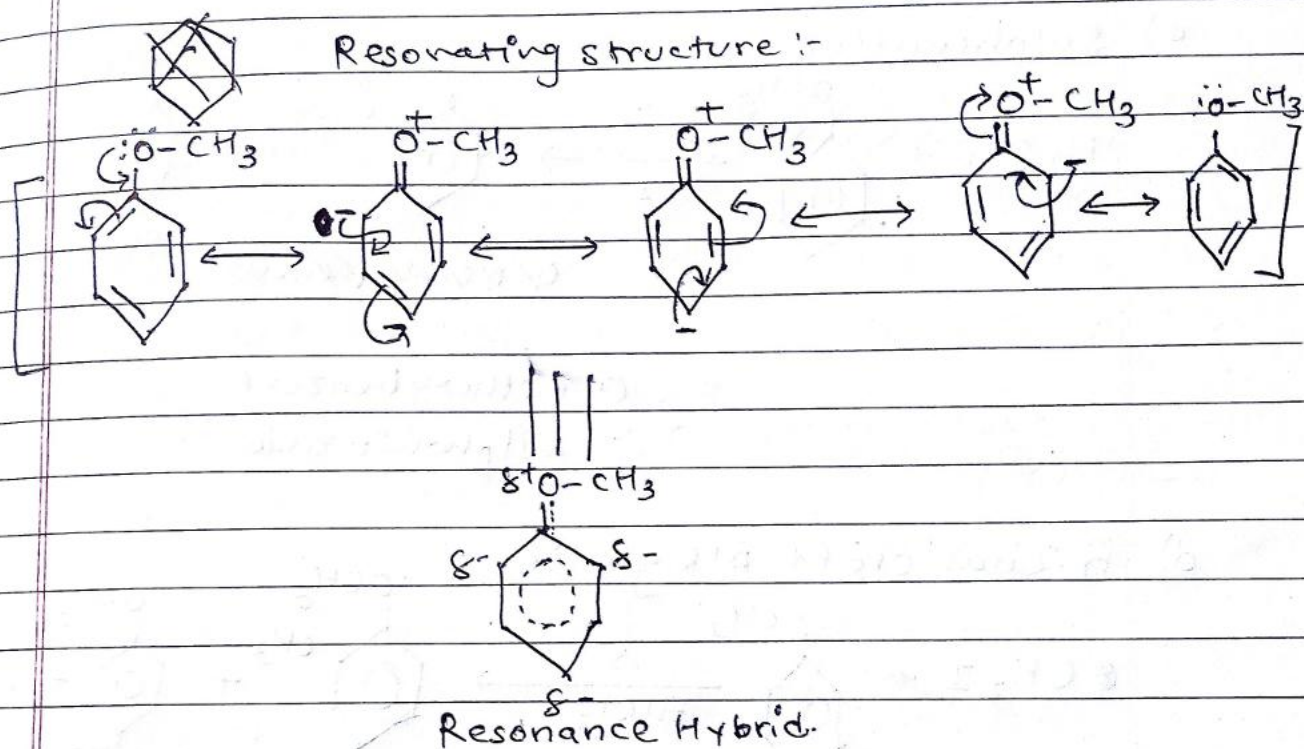
→ Liquid

→ Boiling point $\rightarrow 154^\circ\text{C}$

→ Smells like Anise seed.

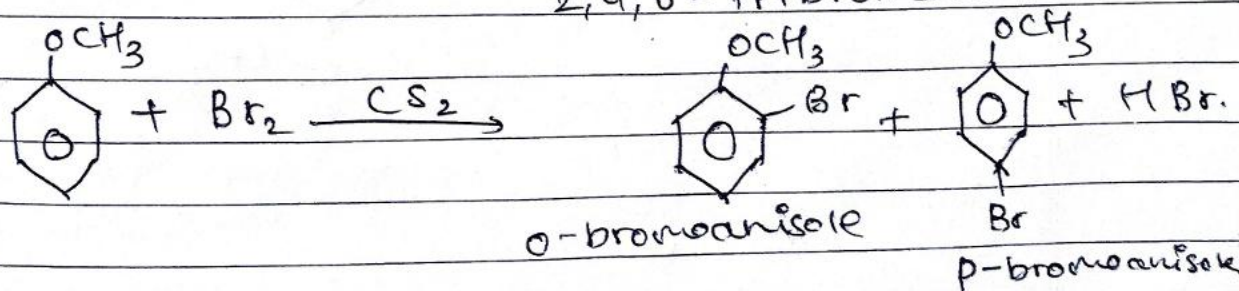
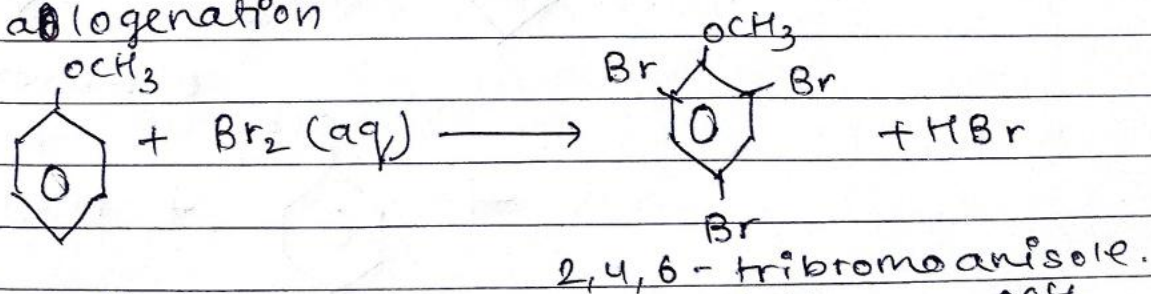
Chemical properties:

① Electrophilic substitution rxn.

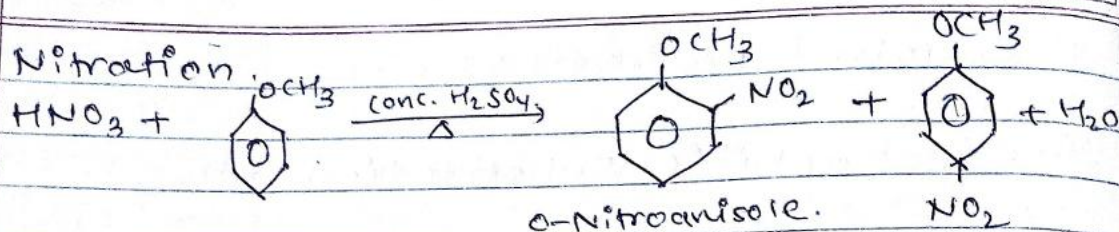


∴ Incoming electrophiles attacks at ortho and para position because they have high electron density than other position (meta).

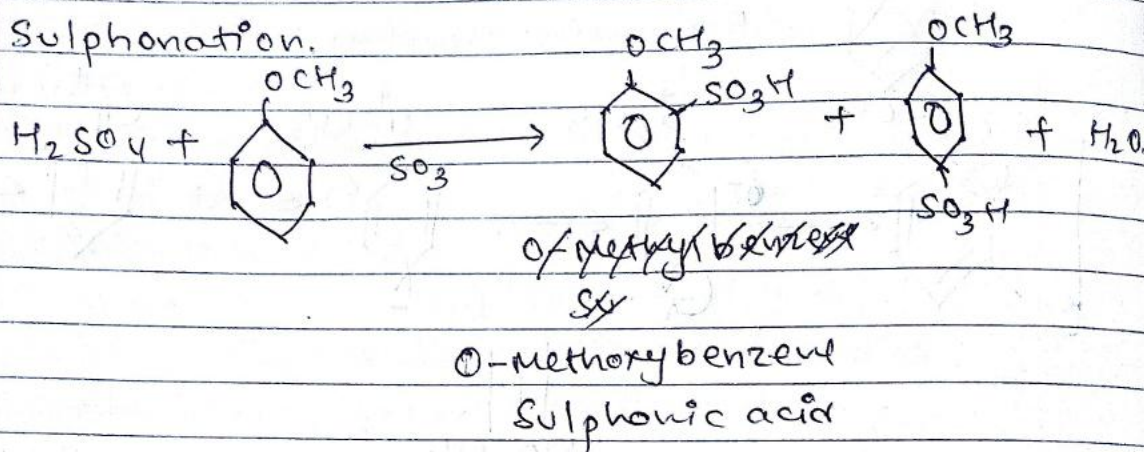
a) Halogenation



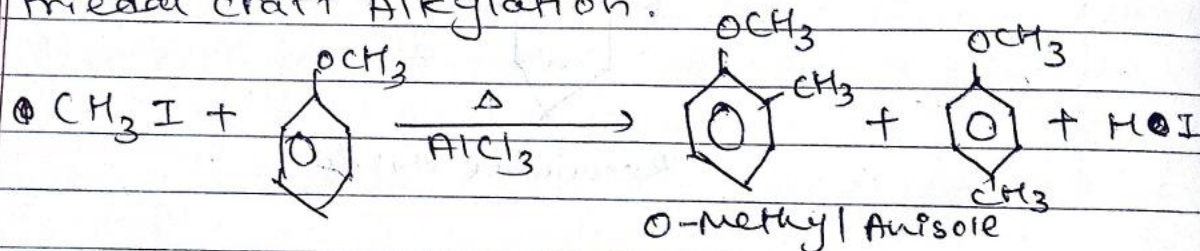
b) Nitration.



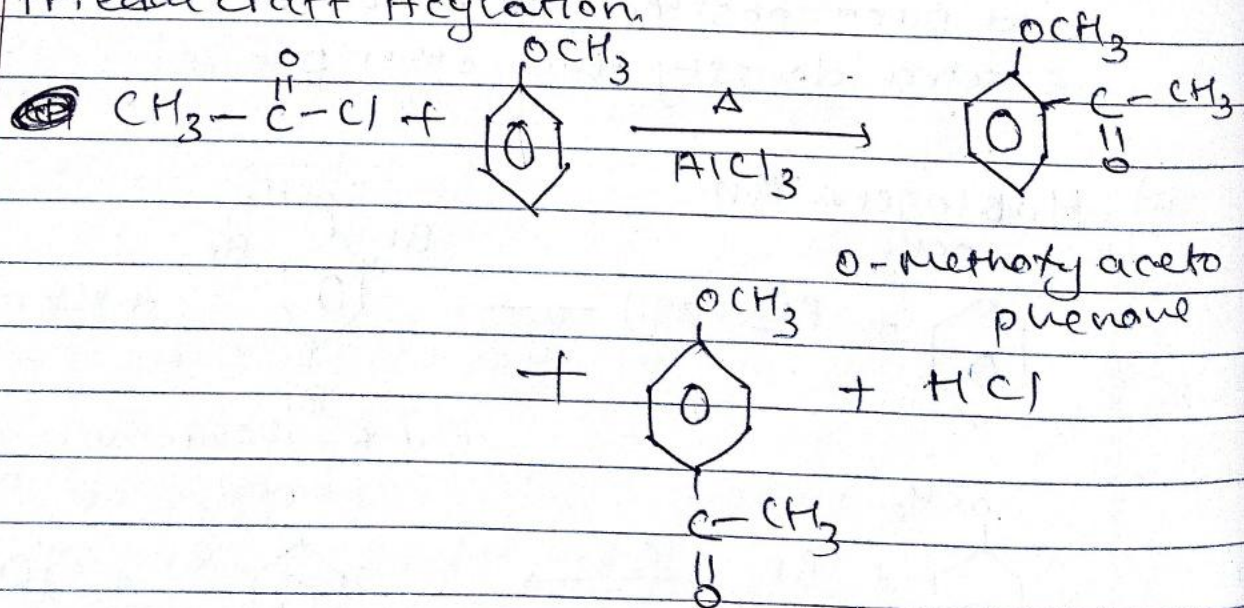
c) Sulphonation.



d) Friedel craft Alkylation.



e) Friedel craft Acylation.



2) Rxn with HI.

